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Automotive drivetrain gearbox including differential and planet gears with blades

Abstract

A drive assembly for a motor vehicle drivetrain includes a differential configured for driving a first output shaft and a second output shaft about a center axis and for allowing the first output shaft and the second output shaft to rotate about the center axis at different speeds, and planet gears each rotatably mounted on a respective planet carrier. The planet gears each include a radially extending surface and blades circumferentially spaced part from each other. The blades axially protrude from the radially extending surface for directing a fluid flow axially toward the differential during operation of the drive assembly.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to motor vehicle drivetrains, and more specifically to differentials in gearboxes for motor vehicle drivetrains.

BACKGROUND

(2) When a differential is supplied with oil via oil sump lubrication and no pump is present to provide oil to the differential of a gearbox, oil supply is difficult, especially at low speeds and high torques.

SUMMARY

(3) A drive assembly for a motor vehicle drivetrain includes a differential configured for driving a first output shaft and a second output shaft about a center axis and for allowing the first output shaft and the second output shaft to rotate about the center axis at different speeds, and planet gears each rotatably mounted on a respective planet carrier. The planet gears each include a radially extending surface and blades circumferentially spaced part from each other. The blades axially protrude from the radially extending surface for directing a fluid flow axially toward the differential during operation of the drive assembly.

- (4) In examples, each of the planet gears includes a disc-shaped base portion rotatably mounted on the respective planet carrier and an outer ring on an outer diameter of the disc-shaped base portion, each outer ring including a plurality of teeth on an outer circumferential surface thereof, the outer ring including the radially extending surface from which the blades axially protrude.
- (5) In examples, the outer ring includes an axially extending surface, the axially extending surface extending from a radially outermost edge of the disc-shaped base portion to the radially extending surface of the outer ring.
- (6) In examples, for each of the planet gears, an axially facing edge of at least one of the blades defines a surface of the planet gear that is axially closest to the differential than any other surface of the planet gear.
- (7) In examples, the differential further includes: a first side gear on a first side of the differential configured for being drivingly connected to the first output shaft; a second side gear on a second side of the differential configured for being drivingly connected to the second output shaft; a differential housing enclosing the first and second side gears; support pins non-rotatably fixed to the differential housing; and spider gears rotatably mounted on the support pins and positioned between the first side gear and the second side gear, each of the spider gears intermeshing with both the first side gear and the second side gear to transmit power from the support pins to the first side gear and the second side gear, the planet carriers drivingly connected to the differential housing to transmit power from the planet gears to the differential.
- (8) In examples, the differential housing includes an axial end that axially faces away from the spider gears, the axial end of the differential housing including axially extending holes extending through differential housing into an interior of the differential housing to supply the fluid flow directed by the blades through the differential housing to the spider gears during operation of the drive assembly.
- (9) In examples, the differential housing includes a differential carrier carrying the support pins, the differential carrier including an annular rim at the axial end including axially extending grooves in an inner circumferential surface of the annular rim partially defining the axially extending holes.
- (10) In examples, the differential housing includes a differential cover connected to the differential carrier and received inside of the annular rim.
- (11) In examples, the differential cover is axially fixed in place in the annular rim by a retainer, the retainer and the axially extending grooves together defining the axially extending holes.
- (12) In examples, the differential cover including a radially extending surface that extends radially inward from an outer diameter of the differential cover, the radially extending surface of the differential cover being axially further away from the blades than a radially extending surface of the annular rim that axially faces the planet gears and that extends from the inner circumferential surface of the annular rim to an outer circumferential surface of the annular rim.
- (13) In examples, the annular rim is axially aligned with the planet carriers.
- (14) In examples, during operation of the drive assembly, the blades follow a path having an inner radial extent that is closer to a center axis of the drive assembly than the inner circumferential surface of the annular rim is to the center axis.
- (15) In examples, the differential housing includes a base section surrounding the spider gears and a flange protruding radially outward from the base section, the planet carriers being non-rotatably fixed to the flange.
- (16) In examples, the drive assembly further includes a sun gear driving the planet gears; and a hollow drive shaft, the sun gear being non-rotatably fixed to an outer diameter of the hollow drive shaft.
- (17) In examples, the first output shaft is received inside of the hollow drive shaft to define a tubular gap configured to direct fluid flow toward the planet gears.
- (18) A gearbox is also provided including the drive assembly; and a gearbox housing defining a sump for receiving oil for lubricating the differential, the differential and the planet gears being

received inside of the gearbox housing.

(19) A method of constructing a drive assembly for a motor vehicle drivetrain is also provided. The method includes inserting a first side gear, a second side gear, support pins and spider gears inside a differential housing; arranging planet gears including blades adjacent to the differential housing, the planet gears being arranged such that during operation of the drive assembly, the blades follow a path having an inner radial extent that is closer to a center axis of the drive assembly than an outer diameter of the differential housing is to the center axis, the spider gears being rotatably mounted on the support pins and positioned between the first side gear and the second side gear, each of the spider gears intermeshing with both the first side gear and the second side gear to transmit power from the support pins to the first side gear and the second side gear for allowing the first side gear and the second side gear to rotate about the center axis at different speeds.

(20) In examples, the method further includes creating axially extending holes in an axial end of the differential housing facing the planet gears, the axially extending holes extending through differential housing into an interior of the differential housing to supply a fluid flow directed by the blades through the differential housing to the spider gears during operation of the drive assembly.

(21) In examples, the differential housing includes a differential cover and a differential carrier, and the inserting of the first side gear, the second side gear, the support pins and the spider gears inside the differential housing includes: inserting the first side gear, the second side gear, the support pins and the spider gears inside the differential carrier; and then connecting the differential cover to the differential carrier, the differential cover forcing the first side gear into the spider gears, the axially extending holes being at an interface between the differential carrier and the differential cover.

(22) In examples, the differential carrier includes an annular rim at the axial end including axially extending grooves in an inner circumferential surface of the annular rim partially defining the axially extending holes, the connecting of the differential cover to the differential carrier including inserting the differential cover into the annular rim and axially fixing the differential cover to the differential carrier via a retainer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is described below by reference to the following drawings, in which:

(2) FIG. 1 shows a radial cross-sectional view of a drive unit of a motor vehicle drivetrain including a drive assembly, which includes a gearbox including a differential, planet gears including blades, and an oil sump;

(3) FIG. 2 shows an enlarged radial cross-sectional view of the gearbox of FIG. 1;

(4) FIG. 3 shows a further enlarged radial cross-sectional view of the gearbox of FIG. 1.

(5) FIG. 4 shows a further radial cross-sectional view illustrating the differential and the planet gears;

(6) FIG. 5 shows a view looking axially into the differential from the position of planet gears;

(7) FIG. 6 also shows a radial cross-sectional of the axial end of the differential housing;

(8) FIG. 7 shows standalone perspective view of the differential carrier, illustrating axially extending grooves on the inner circumferential surface of annular rim; and

(9) FIG. 8 shows standalone radial cross-sectional view of the differential carrier.

DETAILED DESCRIPTION

(10) FIG. 1 shows a radial cross-sectional view of a drive unit **10** of a motor vehicle drivetrain including a drive assembly, which includes a gearbox **12** including a differential **14**, planet gears **16** including blades **18**, and an oil sump **19**; FIG. 2 shows an enlarged radial cross-sectional view of the gearbox **12**; and FIG. 3 shows a further enlarged radial cross-sectional view of the gearbox **12**. The planet gears **16** each include a radially extending surface **16a** and blades **18** circumferentially

spaced part from each other. The blades **18** axially protrude from the radially extending surface **16a** for directing a fluid flow axially toward the differential **14** during operation of the drive assembly. (11) Gearbox **12** further includes a first output shaft **22** and a second output shaft **24** is configured for driving a first axle and a second axle of the motor vehicle drivetrain about a center axis CA and for allowing the first output shaft **22** and the second axle to rotate about the center axis CA at different speeds. The first output shaft **22** is received inside of the hollow drive shaft **26** to define a tubular gap G configured to direct fluid flow F toward the planet gears **16**.

(12) In the example shown in FIGS. **1**, the differential **14** receives a power input from an electric motor **28** via drive shaft **26**, which is a rotor shaft non-rotatably connected a rotor **30** of electric motor **28**. In other example, the input shaft can be driven by an internal combustion engine alone, or in combination with an electric motor when the vehicle is a hybrid. The differential **14** further includes a first side gear **32** on a first side of the differential **14** configured for being drivingly connected to the first output shaft **22** to transmit power from input shaft **26** to the first output shaft **22** for driving a first wheel of the motor vehicle drivetrain.

(13) The differential **14** also includes a second side gear **34** on a second side of the differential **14** configured for being drivingly connected to the second output shaft **24** to transmit power to the first output shaft **22** for driving a second wheel of the motor vehicle drivetrain. Spider gears **36** are rotatably mounted on support pins **38**, which are positioned between the first side gear **32** and the second side gear **34**. Each of the spider gears **36** intermeshes with both the first side gear **32** and the second side gear **34** to transmit power from the support pins **38** to the first side gear **32** and the second side gear **34**. First side gear **32** includes a tubular hub **33** having an inner circumferential surface that includes teeth or splines for non-rotatably fixing first side gear **32** with respect to first output shaft **22**. Second side gear **34** includes a tubular hub **35** having an inner circumferential surface that includes teeth or splines for non-rotatably fixing second side gear **34** with respect to second output shaft **24**.

(14) In the example of FIGS. **1** to **3**, as noted above, drive shaft **26** is the rotor shaft of an electric motor **28**, with drive shaft **26** being non-rotatably fixed to rotor **30** of electric motor **28** and being rotated about center axis CA with rotor **30** by a stator **44** of electric motor **28** in a known manner. Drive shaft **26** is provided with a sun gear **46** non-rotatably fixed to the outer diameter thereof. Sun gear **46** drives planet gears **16**, which in turn drive a differential carrier **50** of differential **14** via planet carriers **52**. Carrier **50** is configured for transmitting power from the planet gears **16** to the support pins **38**. In particular, carrier **50** includes radially extending slots **50d** formed therein, and each of the support pins **38** extends into a respective one of the slots **50d**. Carrier **50** includes a rounded base section **50a**, which curves away from center axis CA, surrounding spider gears **36** and carrying support pins **38**, and a tubular hub section **50b** surrounding tubular hub **35** such that second side gear **34** is rotatable with respect to carrier **50**. On an opposite of carrier **50** as tubular hub section **50b**, carrier **50** includes an annular rim **50c**. Carrier **50** further includes a flange **50e** protruding radially outward from the base section **50b**, and the planet carriers **52** are non-rotatably fixed to the flange **50c**.

(15) Differential housing **20** further includes a differential cover **54** axially fixed to carrier **50** to hold first side gear **32** axially in place inside of carrier **50**. More specifically, cover **54** is received within annular rim **50c** of carrier **50** and is held axially in place by a retainer in the form of snap ring **56** provided in a groove on an inner circumferential surface of annular rim **50c**. The annular rim **50c** is axially aligned with the planet gears **16**. During operation of the drive assembly, the blades **18** follow a path having an inner radial extent that is closer to center axis CA of the drive assembly than the inner circumferential surface of the annular rim **50c** is to the center axis CA. This allows blades **18** to advantageously direct fluid into the annular rim **50c**, and into the interior of the differential housing **20**.

(16) Drive unit **10** includes a housing **58** that is formed by a first end housing section **58a** axially and radially surrounding differential **14** and gears **46**, **16**, an intermediate housing section **58b**

radially surrounding electric motor **28** and a second end housing section **58c** axially enclosing electric motor **28**. Intermediate housing section **58b** includes a radially extending annular wall **58d** defining, together with first end housing section **58a**, a housing of gearbox **12**. The housing of gearbox **12** defines oil sump **19**, which is a lubrication chamber provided within gearbox **12** that is provided with oil for lubricating differential **14** during operation.

(17) Planet gears **16** are each part of a compound planet gear set that further include an auxiliary gear **60** formed as a single piece with the respective planet gear **16**. Each auxiliary gear **60** is axially offset from the respective planet gear **16** and engages with a ring gear **61** that is fixed to housing **58**.

(18) FIGS. **2** and **3** show enlarged view of one of planet gears **16** to illustrate the configuration of blades **18** on each of planet gears **16**. The planet gear **16** includes a radially extending surface **16a** and blades **18** circumferentially spaced part from each other. The blades **18** axially protrude from the radially extending surface **16a** for directing a fluid flow axially toward the differential **14** during operation of the drive assembly.

(19) Each of the planet gears **16** includes a disc-shaped base portion **62** rotatably mounted on the respective planet carrier **52** and an outer ring **64** on an outer diameter of the disc-shaped base portion **62**. Each outer ring **64** including a plurality of teeth **66** on an outer circumferential surface thereof, and further includes the radially extending surface **16a** from which the blades **18** axially protrude. The outer ring **64** includes an axially extending surface **64a** that extends from a radially outermost edge **62a** of the disc-shaped base portion **62** to the radially extending surface **16a** of the outer ring **64**. For each of the planet gears **16**, an axially facing edge **18a** of at least one of the blades **18** defines a surface of the planet gear **16** that is axially closest to the differential than any other surface of the planet gear **16**. Each blade **18** includes two opposite circumferentially facing surfaces **18b**, with one of surfaces **18b** displacing fluid as planet gears **16** are rotated about center axis CA (FIG. **1**). Blades **18** also include an outer diameter surface **18c** and an inner diameter surface **18d**.

(20) FIG. **4** shows a further radial cross-sectional view illustrating the differential **14** and the planet gears **16**. As shown in FIGS. **3** and **4**, a fluid flow **68** generated by blades **18** flows through axially extending holes **70**, which are circumferentially spaced apart from each other, formed in housing **20**. Specifically, referring to FIGS. **1** and **3** together, axially extending holes **70** are formed in an axial end **20a** of differential housing **20** that faces away from spider gears **36**. Holes **70** extend through the differential housing **20** into an interior of the differential housing **20** to supply the fluid flow directed by the blades **18** through the differential housing **20** to the spider gears **36** during operation of the drive assembly. More specifically, annular rim **50c** at the axial end **20a** includes axially extending grooves **70a** in an inner circumferential surface of the annular rim partially defining the axially extending holes **70**.

(21) As noted above, the differential cover **54** of differential housing **20** is axially fixed in place in the annular rim **50c** by a retainer in the form of a snap ring **56**, which forms part of the axially extending holes **70** together with the axially extending grooves **70a**, as shown further in FIG. **5**, which is a view looking axially into from the position of planet gears **16**. In other words, FIG. **5** is an axially facing view in the axial direction of fluid flow **68**. FIG. **6** also shows a radial cross-sectional of the axial end **20a** of differential housing **20**.

(22) As shown in FIG. **6**, the differential cover **54** includes a radially extending surface **54a** that extends radially inward from an outer diameter of the differential cover **54**. The radially extending surface **54a** of the differential cover **54** is axially further away from the blades **18** than a radially extending surface **20b** of the annular rim **50c** that axially faces the planet gears **16**. Surface **20b** extends from the inner circumferential surface of the annular rim **50c** to an outer circumferential surface of the annular rim **50c**.

(23) FIGS. **7** and **8** show standalone perspective view and radial cross-sectional views, respectively, of the differential carrier **50**, illustrating axially extending grooves **70a** on the inner circumferential

surface of annular rim **50c** and an annular groove **72** for receiving the snap ring **56**.

(24) In the preceding specification, the disclosure has been described with reference to specific exemplary embodiments and examples thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of disclosure as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative manner rather than a restrictive sense.

REFERENCE NUMERALS

(25) **10** drive unit **12** gearbox **14** differential **16** planet gears **16a** radially extending surface **18** blades **18a** axially facing edge **18b** two opposite circumferentially facing surfaces **18c** outer diameter surface **18d** inner diameter surface **19** oil sump **20** differential housing **20a** axial end **20b** radially extending surface **22** first output shaft **24** second output shaft **26** drive shaft **28** electric motor **30** rotor **32** first side gear **33** tubular hub **34** second side gear **35** tubular hub **36** spider gears **38** support pins **44** stator **46** sun gear **50** carrier **50a** rounded base section **50b** tubular hub section **50c** annular rim **50d** slots **50e** flange **52** planet carriers **54** differential cover **54a** radially extending surface **56** snap ring **58** housing **58a** first end housing section **58b** intermediate housing section **58c** second end housing section **58d** radially extending annular wall **60** auxiliary gear **61** ring gear **62** disc-shaped base portion **62a** radially outermost edge **64** outer ring **64a** axially extending surface **66** plurality of teeth **68** fluid flow **70** axially extending holes **70a** axially extending grooves **72** annular groove

Claims

1. A drive assembly for a motor vehicle drivetrain, the drive assembly comprising: a differential configured for driving a first output shaft and a second output shaft about a center axis and for allowing the first output shaft and the second output shaft to rotate about the center axis at different speeds; and planet gears each rotatably mounted on a respective planet carrier, the planet gears each including a radially extending surface and blades circumferentially spaced part from each other, the blades axially protruding from the radially extending surface for directing a fluid flow axially toward the differential during operation of the drive assembly.
2. The drive assembly as recited in claim 1 wherein each of the planet gears includes a disc-shaped base portion rotatably mounted on the respective planet carrier and an outer ring on an outer diameter of the disc-shaped base portion, each outer ring including a plurality of teeth on an outer circumferential surface thereof, the outer ring including the radially extending surface from which the blades axially protrude.
3. The drive assembly as recited in claim 2 wherein the outer ring includes an axially extending surface, the axially extending surface extending from a radially outermost edge of the disc-shaped base portion to the radially extending surface of the outer ring.
4. The drive assembly as recited in claim 3 wherein, for each of the planet gears, an axially facing edge of at least one of the blades defines a surface of the planet gear that is axially closest to the differential than any other surface of the planet gear.
5. The drive assembly as recited in claim 2 wherein the differential includes: a first side gear on a first side of the differential configured for being drivingly connected to the first output shaft; a second side gear on a second side of the differential configured for being drivingly connected to the second output shaft; a differential housing enclosing the first and second side gears; support pins non-rotatably fixed to the differential housing; and spider gears rotatably mounted on the support pins and positioned between the first side gear and the second side gear, each of the spider gears intermeshing with both the first side gear and the second side gear to transmit power from the support pins to the first side gear and the second side gear, the planet carriers drivingly connected to the differential housing to transmit power from the planet gears to the differential.
6. The drive assembly as recited in claim 5 wherein the differential housing includes an axial end

that axially faces away from the spider gears, the axial end of the differential housing including axially extending holes extending through differential housing into an interior of the differential housing to supply the fluid flow directed by the blades through the differential housing to the spider gears during operation of the drive assembly.

7. The drive assembly as recited in claim 6 wherein the differential housing includes a differential carrier carrying the support pins, the differential carrier including an annular rim at the axial end including axially extending grooves in an inner circumferential surface of the annular rim partially defining the axially extending holes.

8. The drive assembly as recited in claim 7 wherein the differential housing includes a differential cover connected to the differential carrier and received inside of the annular rim.

9. The drive assembly as recited in claim 8 wherein the differential cover is axially fixed in place in the annular rim by a retainer, the retainer and the axially extending grooves together defining the axially extending holes.

10. The drive assembly as recited in claim 8 wherein the differential cover including a radially extending surface that extends radially inward from an outer diameter of the differential cover, the radially extending surface of the differential cover being axially further away from the blades than a radially extending surface of the annular rim that axially faces the planet gears and that extends from the inner circumferential surface of the annular rim to an outer circumferential surface of the annular rim.

11. The drive assembly as recited in claim 7 wherein the annular rim is axially aligned with the planet carriers.

12. The drive assembly as recited in claim 11 wherein during operation of the drive assembly, the blades follow a path having an inner radial extent that is closer to a center axis of the drive assembly than the inner circumferential surface of the annular rim is to the center axis.

13. The drive assembly as recited in claim 5 wherein the differential housing includes a base section surrounding the spider gears and a flange protruding radially outward from the base section, the planet carriers being non-rotatably fixed to the flange.

14. The drive assembly as recited in claim 1 further comprising: a sun gear driving the planet gears; and a hollow drive shaft, the sun gear being non-rotatably fixed to an outer diameter of the hollow drive shaft.

15. The drive assembly as recited in claim 14 wherein the first output shaft is received inside of the hollow drive shaft to define a tubular gap configured to direct fluid flow toward the planet gears.

16. A gearbox comprising: the drive assembly as recited in claim 1; a gearbox housing defining a sump for receiving oil for lubricating the differential, the differential and the planet gears being received inside of the gearbox housing.
